2022-Summer Paper 4 – Recent Advances

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Q1. Natural orifice endoscopic surgery.

Type: LAQ (25 marks)

Taken as NOTES (natural orifice transluminal endoscopic surgery) and its relatives (hybrid NOTES, LESS [laparoendoscopic single-site surgery]).

Definition and concept

- **NOTES:** entry into the peritoneal or retroperitoneal cavity through a deliberate incision in a hollow viscus reached through a natural orifice (mouth, vagina, bladder, rectum). The aim is **"scarless" surgery**: no abdominal wall wound, so no wound pain, infection or hernia.
- It differs from **endoluminal** surgery (e.g. ureteroscopy), which stays inside the lumen.
- **Pure NOTES:** all instruments through the orifice. **Hybrid NOTES:** natural orifice plus one or more small abdominal ports. Hybrid is what is actually used in humans.
- Chapter 142 places NOTES at the end of a spectrum: transperitoneal → retroperitoneal → hand-assisted → LESS → SP (single-port) robotic → NOTES. **Each step to fewer incisions buys cosmesis and less pain at the cost of working space and triangulation.**

History (milestones)

Year	Event
2002	Transvaginal laparoscopic nephrectomy in pigs (Gettman, Cadeddu)
2004	Transgastric peritoneoscopy in pigs (Kalloo)
2006	NOSCAR (Natural Orifice Surgery Consortium for Assessment and Research) white paper listed barriers
2008–2010	Hybrid, then pure, transvaginal nephrectomy in humans

Routes of access

Route	Features	Urological use
Transvaginal (posterior colpotomy)	Commonest; easy closure under vision; large specimens pass; women only	Nephrectomy, donor nephrectomy, specimen extraction
Transgastric	Upper abdomen reachable; gastrotomy closure and leak are the main risk	Experimental (peritoneoscopy, renal surgery in animals)

Route	Features	Urological use
Transvesical	Sterile urine; bladder dome port; small calibre; closure difficult	Peritoneoscopy (animal, Lima), transvesical access to pelvis
Transurethral	Urologist's familiar route	Experimental transurethral prostatectomy in cadavers
Transcolonic/transanal	Faecal contamination risk	Rarely urological

Hybrid NOTES and natural orifice extraction

- Standard laparoscopy does the dissection; **the vagina is the access for a camera or instrument and the extraction route** (Chapter 142). This is the only NOTES form with real clinical urological use.
- **NOSE (natural orifice specimen extraction):** kidney removed through the vagina after laparoscopic nephrectomy, avoiding the extraction incision, which is the largest and most painful wound.
- **NOTES-assisted donor nephrectomy:** laparoscopic or robotic donor nephrectomy with **transvaginal kidney extraction**. Reported benefits: less pain, faster recovery, no extraction scar; warm ischaemia acceptable.

Steps of transvaginal (hybrid) nephrectomy

1. Lithotomy then modified flank; bowel and vaginal preparation; antibiotics.
2. Umbilical port (camera or instrument) plus one or two small ports.
3. **Posterior colpotomy** under laparoscopic vision; long port or flexible scope through the vagina.
4. Colon reflection, hilar dissection, ureter and vessels stapled.
5. Kidney in a bag, delivered through the colpotomy.
6. Colpotomy closed with absorbable sutures under vision.

LESS: the close relative

- All ports through **one skin incision** (umbilicus or below the waistline) using several low-profile trocars or a multichannel port (TriPort; da Vinci SP 25–27-mm entry guide).
- Problems: **no triangulation**, instrument clashing, in-line view. Solutions: articulating or bent instruments, flexible-tip scopes, instruments of different lengths.
- Evidence (Chapter 142): randomised trial versus laparoscopy showed the same time, blood loss and complications, with less analgesia. Donor meta-analysis: lower pain scores. Oncological outcome unchanged.
- Drawback (Chapter 20): **more incisional and umbilical hernias** because of the larger single incision.
- **SP robot (FDA [Food and Drug Administration] approval 2018)** has revived single-site surgery.

vNOTES (vaginal NOTES)

- A gynaecological development: a single-port device in the vagina gives access to the peritoneum.
- **HALON RCT (randomised controlled trial):** vNOTES hysterectomy was non-inferior to laparoscopic hysterectomy, with more same-day discharge and less pain.
- Urogynaecological uses: vNOTES apical prolapse repair (uterosacral suspension), adnexal surgery.

Challenges (NOSCAR list)

- **Safe access** and **secure closure** of the viscerotomy (leak, fistula); peritoneal **infection**.

- **Spatial orientation**: retroflexed, off-axis view.
- **Platform**: flexible scopes lack stiffness, traction and triangulation.
- Control of **haemorrhage** and other complications.
- Training and cost; dyspareunia and fertility concerns after colpotomy; transvaginal route unavailable in men.
- Chapter 142: adoption is limited by **no clear therapeutic benefit**, safety concerns and very selective patients; **pure NOTES has very few urological cases.**

Recent advances

- **Flexible robotic platforms** (Chapter 39): MASTER (NOTES, haptics), ViaCath.
- **Transvesical single-port** (LESS or da Vinci SP) surgery for diverticulectomy and foreign bodies (Chapter 124).
- Magnetic anchoring and guidance systems to restore triangulation.

Also know: LESSCAR consortium standardised the term LESS; "embryonic NOTES" (umbilicus as the natural scar); hidden-incision endoscopic surgery (HIdES) in children.

Diagram to draw: sagittal female pelvis showing posterior colpotomy with a transvaginal port and kidney extraction, plus umbilical port.

> **Key points to remember:**
> - NOTES = transluminal via natural orifice; hybrid transvaginal is the only routinely used urological form.
> - Transvaginal extraction removes the most painful wound.
> - LESS: one incision, lost triangulation, more hernias.
> - Barriers: closure, infection, orientation, platform; no proven benefit yet.

Q2. Tissue engineering in urology.

Type: LAQ (25 marks)

Definition and rationale

- **Tissue engineering:** applying engineering and life sciences to build biological substitutes that restore, maintain or improve tissue function.
- **Why in urology:** bowel used for bladder and diversion causes mucus, stones, metabolic acidosis, infection and malignancy; grafts for urethra are limited in supply. The aim is **to replace "wrong tissue" with "right tissue"**.
- **Regenerative medicine** is the wider term: cells, scaffolds, growth factors and gene therapy.

The tissue-engineering triad

```
Cells  +  Scaffold  +  Signals (growth factors, mechanical cycling in a bioreactor)
          -> construct -> implanted -> vascularised and remodelled by the host
```

Cell sources

Source	Features
Autologous differentiated cells (urothelial and smooth muscle cells from a bladder biopsy)	No rejection; expanded in culture; unusable if the bladder is malignant or diseased
Adult stem cells: bone marrow or adipose mesenchymal, muscle-derived, urine-derived	Easy harvest; paracrine effect
Embryonic stem cells	Pluripotent; ethics, teratoma risk
iPSCs (induced pluripotent stem cells)	Adult cells reprogrammed; patient-specific; tumour risk

Scaffolds (biomaterials)

- **Requirements:** biocompatible; biodegradable at the rate of tissue ingrowth; porous for cell attachment and vessel ingrowth; mechanically suitable (compliance for bladder).

Type	Examples	Comment
Natural	Collagen, alginate, silk fibroin	Good cell recognition, weaker
Acellular tissue matrices	Bladder acellular matrix, SIS (small intestinal submucosa), decellularised dermis	Retain native architecture and growth factors
Synthetic polymers	PGA (polyglycolic acid), PLA (polylactic acid), PLGA (their copolymer)	Controlled degradation and shape; no bioactivity
Newer	Electrospun nanofibres, hydrogels	Mimic extracellular matrix

- **Unseeded (acellular)** grafts depend on host cell migration and work only for short or onlay defects (about 0.5 cm for tubular urethra in animal work). Longer gaps need **seeded** scaffolds, or they fibrose and contract.

Clinical applications

1. Bladder

- **Atala et al. (2006):** first seeded biodegradable bladder construct (autologous urothelial and muscle cells on collagen–PGA, wrapped in omentum) for augmentation in **7 myelomeningocele patients**; compliance improved.

- **Industry (Tengion) multi-institution phase II trial** in spina bifida: **no clinically or statistically significant improvement in compliance or capacity** (Chapter 48).
- Chapter 48 conclusion: a bladder with specialised urothelium and contractile muscle has **not** been achieved; it remains experimental.
- Chapter 111: an engineered bladder gives **storage only**; innervation for voluntary emptying cannot be recreated.

2. Urinary diversion: Neo-Urinary Conduit (Tengion, NCT01087687) (Chapter 135)

- Seeded biodegradable conduit to avoid bowel. Small enrolment; conduits survived about 250 days on average; **half developed stenosis or stricture.**
- Lessons: stiff, non-collapsible material; no cells formed around the connective-tissue sleeve; **no blood supply**.

3. Urethra

- Acellular bladder matrix onlay for hypospadias and stricture: success in short onlay repairs.
- **Raya-Rivera et al. (2011, Lancet):** tubular PGA–PLGA scaffolds seeded with autologous cells in **5 boys** with posterior urethral defects; functional urethras at up to 6 years.
- Tissue-engineered oral mucosa (MukoCell) for urethroplasty, avoiding donor-site morbidity.

4. Other organs

- **Sphincter:** autologous muscle-derived cells for SUI (stress urinary incontinence); also for underactive detrusor (Chapter 111: open-label improvement). Biodegradable scaffold slings (Chapter 48).
- **Vagina:** engineered vaginas in MRKH (Mayer–Rokitansky–Küster–Hauser) syndrome (Raya-Rivera, 2014).
- **Penis:** engineered corporal tissue, animal stage. ED (erectile dysfunction) stem cell therapy is investigational (Chapter 72).
- **Kidney:** decellularised–recellularised scaffolds (animal); renal assist devices; kidney organoids.
- **Testis:** spermatogonial stem cell banking for prepubertal cancer patients (future, Chapter 4).

3D bioprinting

- Layer-by-layer deposition of **"bio-ink"** (cells in hydrogel) with or without polymer, driven by patient CT (computed tomography) or MRI (magnetic resonance imaging) data.
- Advantages: exact patient-specific shape, multiple cell types placed precisely, built-in channels for vessels.
- Current urological use: 3D-printed (non-living) kidney and prostate models for planning and teaching; bioprinted urethra and bladder constructs are still at the animal stage.

Organoids

- 3D self-organising mini-organs grown from stem cells or tumour tissue.
- **Kidney organoids** from iPSCs: models of development and disease, drug toxicity testing.
- **Tumour organoids** (bladder, prostate): personalised drug testing and biology (WNT/β-catenin work, Chapter 129).

Challenges

- **Vascularisation** of thick constructs (diffusion limit), **innervation** and contractility.
- Fibrosis and contracture; scale-up, cost, culture time; regulatory approval.
- Safety of stem cells (tumour formation). Disappointing controlled trials after promising pilots.

Recent advances

- Robotic bladder auto-transplantation and vascularised composite bladder allograft (Chapter 135): a transplant, not engineered, route to "right tissue".
- Smart scaffolds releasing growth factors; urine-derived stem cells; gene-edited cells.

Also know: bioreactors apply cyclic stretch to train smooth muscle.

Diagram to draw: biopsy → cell culture → seeding on scaffold → bioreactor → implant with omental wrap.

> **Key points to remember:**
> - Triad: cells, scaffold, signals; vascularity is the main barrier.
> - Atala 2006 (7 patients) promising; phase II trial: no gain in compliance or capacity.
> - Neo-urinary conduit: half stenosed; no blood supply.
> - Raya-Rivera 2011: seeded tubular urethras in 5 boys.

Q3a. Urological applications in nanotechnology.

Type: SAQ (10 marks)

Taken as "applications of nanotechnology in urology".

Definition

- **Nanotechnology:** design and use of materials of about **1–100 nm**. At this scale, surface area and optical, magnetic and thermal properties change.
- **Why useful:** particles pass through leaky tumour vessels (**enhanced permeability and retention**), carry drugs or tracers, and can be targeted with ligands such as PSMA (prostate-specific membrane antigen) antibodies.

Applications

Area	Application
Imaging and staging	**USPIO (ultrasmall superparamagnetic iron oxide) lymphotropic MRI (magnetic resonance imaging):** normal nodes take up particles and lose T2 signal; metastatic nodes do not (Chapter 6; investigational in penile cancer, Chapter 84). Gold-nanoparticle CT (computed tomography) contrast, quantum dots
Sentinel node	Technetium-99m **nanocolloid** for dynamic sentinel node biopsy in penile cancer (Chapter 84)
Diagnosis	Nanowire and nanoparticle biosensors detecting PSA (prostate-specific antigen) and urinary markers at very low levels
Drug delivery	Nanoparticle (albumin-bound) paclitaxel in urothelial and prostate cancer (Chapter 164: tolerated, no pathological complete response); **NanoPac** intratumoral paclitaxel (Chapter 163); liposomal drugs; mucoadhesive intravesical carriers; siRNA (small interfering RNA) delivery
Focal ablation	**Gold nanoshells (AuroLase):** collect in tumour vessels and are heated by near-infrared laser; about three-quarters success on 12-month biopsy (Chapter 163). Magnetic nanoparticle hyperthermia
Irreversible electroporation	Electric pulses create **nanoscale membrane pores** ("NanoKnife"); spares matrix, vessels and ducts; investigational (Chapter 143)
Devices	Silver and other nanocoatings on catheters and stents against biofilm and encrustation
Tissue engineering	Electrospun nanofibre scaffolds that mimic extracellular matrix
Stone disease	"Calcifying nanoparticles" (nanobacteria) linked to Randall plaque, not proven causal (Chapter 95); nano-CT of stones and tissue

Limitations

- Toxicity and long-term tissue accumulation; uptake by liver and spleen; manufacturing cost; regulatory hurdles. Most uses remain experimental.

Diagram to draw: nanoparticle with drug core, polymer shell and surface PSMA ligand entering a leaky tumour vessel.

> **Key points to remember:**
> - 1–100 nm; enhanced permeability and retention allows tumour targeting.
> - USPIO MRI: benign nodes lose signal; metastatic nodes do not.
> - AuroLase gold nanoshells: photothermal focal therapy.

Q3b. Guidelines for declaration of brain stem death.

Type: SAQ (10 marks) (also answered in 2018-Summer Paper 4, Q3c)

Definition and law

- **Brain-stem death:** irreversible loss of consciousness and of the capacity to breathe; equivalent to death of the person.
- **India:** THOA (Transplantation of Human Organs Act) 1994; amended 2011; **Rules 2014**. Internationally: Harvard criteria (1968), AAN (American Academy of Neurology) criteria (updated 2023).

Board of Medical Experts (India)

Member
1. Registered medical practitioner **in charge of the hospital**
2. **Independent specialist** nominated from a panel approved by the Appropriate Authority
3. **Neurologist or neurosurgeon** (if unavailable: surgeon, physician, anaesthetist or intensivist)
4. Medical practitioner **treating the patient**

- **No member may belong to the transplant team.**

Preconditions

- Apnoeic coma on a ventilator with a **known irreversible cause**.
- Exclude **sedatives, relaxants**, hypothermia, metabolic or endocrine causes, shock.

Tests: all reflexes absent

Brainstem reflex	Method	Cranial nerves
Pupillary	Fixed pupils, no light reaction	II, III
Corneal	Touch cornea: no blink	V, VII
Oculocephalic	Turn head: eyes move with head	VIII, III, VI
Oculovestibular	Ice-cold water in each ear: no eye movement	VIII, III, VI
Motor in cranial nerve area	Supraorbital pressure: no grimace	V, VII
Gag and cough	Pharynx touch and tracheal suction: no response	IX, X

- **Apnoea test (last):** pre-oxygenate with 100% oxygen; disconnect ventilator with oxygen insufflation; **no breathing effort at $PaCO_2$ (arterial carbon dioxide tension) ≥60 mmHg** (or ≥20 mmHg above baseline). Abort for hypotension, desaturation or arrhythmia.
- **Two sets of tests at least 6 hours apart**, recorded on **Form 10**.
- **Ancillary tests** if examination is impossible (facial trauma): angiography, perfusion scan, EEG (electroencephalogram).

After declaration

- **Mandatory request** for donation by ICU (intensive care unit) staff; next-of-kin consent; **NOTTO** (National Organ and Tissue Transplant Organisation) allocates organs.

Diagram to draw: flowchart: coma → preconditions → test 1 → 6 h → test 2 → Form 10 → consent → NOTTO.

> **Key points to remember:**
>
> - Four-member Board; no transplant-team member.
> - Two tests ≥6 h apart; Form 10.
> - Apnoea test: no breathing at $PaCO_2$ ≥60 mmHg.

Q3c. Different types of trocars in laparoscopy.

Type: SAQ (10 marks)

Definition

- A **trocar** is a **cannula (port) plus an obturator**; 3–20 mm diameter, 5–20 cm long (Chapter 18).
- Parts: obturator tip, cannula, **valve** (keeps gas in), insufflation side-arm, retention mechanism.

Classification

Basis	Types
Obturator tip	Bladed (cutting) with spring safety shield; **bladeless (blunt dilating)**; **optical (visual)**; blunt (Hasson)
Material	Reusable metal; disposable plastic
Valve	Flap or trumpet valve (needs reducer); **multiseal** (5–12 mm, no reducer); **valveless (AirSeal)**
Retention	Smooth (suture-anchored); threaded sleeve; expandable arms; **balloon (Blunt Tip)**
Special	Hasson open cannula; radially expanding; hand ports; single-site multichannel ports; robotic 8-mm metal ports

Main types explained

- **Bladed trocars:** cut through tissue; more epigastric injury and port-site hernia. Now of historical interest.
- **Bladeless (dilating) trocars:** spread muscle fibres. About **fivefold less inferior epigastric injury**; trocar-site bleeding 3% vs 9%. Switching cut port hernia from 1.83% to 0.19%. **Fascial closure only for midline ports >10 mm** or stretched sites.
- **Optical trocars:** clear tip with a 0° scope inside; each layer is seen during entry.
- **Hasson cannula:** blunt obturator and cone, fixed with fascial stay sutures after open entry; safer with adhesions; leaks more gas.
- **Radially expanding trocar:** a sleeve over the Veress needle is dilated to 12 mm; less port-site bleeding and pain (Chapter 39).
- **Balloon (Blunt Tip) trocar:** internal balloon and external foam cuff seal the wall; prevents leak and **subcutaneous emphysema** in retroperitoneoscopy.
- **AirSeal valveless port:** stable pneumoperitoneum at 8–12 mmHg despite suction.
- **Single-site ports:** TriPort; da Vinci SP (single-port) 25–27-mm guide. **Hand ports:** GelPort.

Safety points

- **Never use a metal trocar with a plastic retaining ring**: capacitive coupling burns viscera out of view.
- Insert under vision; ports 8–10 cm apart.

Also know: Veress needle (14-gauge, spring-loaded blunt stylet) is an entry needle, not a trocar.

Diagram to draw: trocar with obturator tip types (bladed, bladeless, optical) and valve.

> **Key points to remember:**
> - Trocar = cannula + obturator.
> - Bladeless: fewer epigastric injuries and hernias.
> - Balloon trocar prevents retroperitoneal emphysema.

Q3d. Current ureter scopes.

Type: SAQ (10 marks)

Taken as "current ureteroscopes"; see also 2019-Summer Paper 4, Q3c.

Classification

Type	Key features
Rigid rod-lens	Larger calibre; largely historical
Semirigid	**Fibreoptic** image bundle; tip **≤7 Fr**; channel >3 Fr; often **two channels**; in-line or offset eyepiece
Flexible fibreoptic	Tip 4.5–7.5 Fr (smallest); image degrades as fibres break (black dots)
Flexible digital	"Chip-on-tip" CMOS (complementary metal-oxide semiconductor) or CCD (charge-coupled device) sensor plus LED (light-emitting diode); no camera head
Single-use flexible	Digital; LithoVue (2016) first; now several makers
Ureteral resectoscope	11.5 Fr; largely replaced by laser

Semirigid ureteroscope

- Uses **coherent fibre bundles** for the image (random fibres carry only light).
- **Two channels** are preferred: irrigation through one while an instrument is in the other, or a laser through one to break a stone held in a basket.
- **Offset eyepiece** gives a straight channel for rigid probes (pneumatic, biopsy).
- **Rule:** semirigid below the iliac vessels, flexible above.

Flexible ureteroscope

- Parts: optics, deflection mechanism (lever pulls wires), working channel.
- **Working channel 3.6 Fr**; active deflection **>270° both ways** in one plane; enhanced primary deflection reaches the lower pole (ureteroinfundibular angle up to about 175°).

Digital versus fibreoptic

- Digital: sharper image, **shorter operating time, same stone-free rate**. But larger tip, so **more access sheaths and sheath complications**.

Single-use scopes

- **LithoVue:** CMOS, LED, tip 7.7 Fr, shaft 9.5 Fr, 3.6 Fr channel, 270°.
- Reason: reusable scopes may need repair after 3–5 cases.
- Best value in low-volume centres and for **large lower-pole stones**.

Care

- **Fire laser only when the fibre tip is seen touching the stone**; pass fibre with the scope straight. Sterilise with ethylene oxide, glutaraldehyde or peracetic acid.

Recent advances

- Intrarenal **pressure sensing**; AI (artificial intelligence); robotic flexible ureteroscopy; suction access sheaths.

Diagram to draw: flexible scope tip showing CMOS chip, LEDs, 3.6 Fr channel and 270° deflection.

> **Key points to remember:**
> - Semirigid: two channels; below the iliac vessels.
> - Flexible: 3.6 Fr channel, >270° deflection.
> - Digital: shorter operation, larger tip; LithoVue first single-use (2016).

Q3e. General complications related to laparoscopy.

Type: SAQ (10 marks) (adapted from 2019-Summer Paper 1, Q2)

Incidence

- Historically about 4%, mortality 0.03–0.08%. Contemporary series 13–22% because procedures are complex. **Vascular, then adjacent-organ injuries are commonest.**
- Risk factors: complex procedures, operating time >4 h, creatinine >1.5 mg/dL, learning curve.

Complications by phase

Phase	Complication	Key points
Access (Veress, first trocar)	Preperitoneal insufflation; vessel or bowel puncture; **major vascular injury (0.11–2%)**	Confirm Veress tests; open access with adhesions. Major vessel: leave trocar in, laparotomy
Insufflation	**Gas embolism**; subcutaneous emphysema; pneumothorax (1.6–4%); pneumomediastinum, pneumopericardium; hypercarbia	Embolism: end-tidal CO_2 (carbon dioxide) rises then falls, mill-wheel murmur → stop gas, left lateral head-down, 100% oxygen, aspirate central line
Secondary trocars	Epigastric bleeding; crossing swords, rollover	Transilluminate; bladeless trocars; insert under vision
Dissection	Thermal or mechanical bowel injury; vascular injury; bladder or ureter injury	Four electrosurgical mechanisms below
Positioning	Nerve injury (about 2.8%: brachial plexus, femoral, sciatic); **rhabdomyolysis** (about 1% retroperitoneal)	Padding, kidney rest <1 h, cases <5 h
Exit	Bowel entrapment, late port bleeding	Inspect each site at 5 mmHg
Postoperative	Port-site hernia (0.3–5.4%), wound infection, shoulder-tip pain, DVT (deep vein thrombosis)	Close fascia of large ports

Electrosurgical bowel injury: four mechanisms

1. **Inappropriate activation** out of view.
2. **Direct coupling** with another metal instrument.
3. **Capacitive coupling** (metal cannula in a plastic retainer).
4. **Insulation failure.**

- Monopolar injury presents **late (up to 18 days)**; resect with margin.

Missed bowel injury (Bishoff)

- Not classic peritonitis: **pain at one trocar site, leucopenia, low fever, distension, diarrhoea** at 24–48 h.
- **CT (computed tomography) with oral contrast**; free gas is normal for up to 7–9 days.

Physiological complications

- Hypercarbia and acidosis; oliguria from renal vein compression; reduced venous return; raised intracranial pressure in head-down tilt.

Diagram to draw: abdominal wall showing capacitive coupling through a metal cannula in a plastic anchor.

> **Key points to remember:**
>
> - Vascular injury is commonest; leave the trocar in place.
> - Gas embolism: left lateral, head down.
> - Missed bowel injury: trocar-site pain, leucopenia.

Q3f. Approach to evaluation of adrenal mass/s.

Type: SAQ (10 marks)

Introduction

- **Adrenal incidentaloma:** mass ≥1 cm found on imaging done for another reason.
- Two questions: **Is it malignant? (imaging)** and **Is it functioning? (biochemistry).**

Differential diagnosis

- Non-functioning adenoma (commonest); cortisol-secreting adenoma (5.3%); phaeochromocytoma (5.1%); aldosteronoma (1%); ACC (adrenocortical carcinoma, 4.7%); metastasis (2.5%); myelolipoma, cyst, haemorrhage.
- **Bilateral:** metastases, lymphoma, **tuberculosis** and histoplasmosis, haemorrhage, CAH (congenital adrenal hyperplasia), macronodular hyperplasia, bilateral phaeochromocytoma (MEN-2 [multiple endocrine neoplasia type 2], VHL [von Hippel–Lindau]).

Imaging

- CT (computed tomography), HU (Hounsfield units), MRI (chemical-shift magnetic resonance imaging), MDT (multidisciplinary team), FDG-PET (fluorodeoxyglucose positron emission tomography).

```
Unenhanced CT -> <=10 HU or MRI signal drop-out -> lipid-rich adenoma
              -> >10 HU -> washout CT (15 min): APW >60% or RPW >40% -> adenoma
                          -> indeterminate -> MDT: repeat imaging 6-12 months, FDG-PET or
surgery
```

- **APW** (absolute percentage washout) = (enhanced − delayed) ÷ (enhanced − unenhanced) × 100; **RPW** (relative) = (enhanced − delayed) ÷ enhanced × 100.
- Malignant features: heterogeneous, irregular, invasion, >4 cm. Macroscopic fat = myelolipoma.
- **Size and malignancy:** <4 cm 0.1%; 4–6 cm 2.4%; >6 cm 19.5%.

Hormonal work-up (all lesions)

Hormone	Test
Cortisol	**1-mg overnight DST (dexamethasone suppression test):** ≤1.8 µg/dL (50 nmol/L) excludes; >1.8 = **MACS** (mild autonomous cortisol secretion); >5 µg/dL autonomous
Catecholamines	**Plasma free or 24-h urinary fractionated metanephrines**
Aldosterone	**ARR** (aldosterone-to-renin ratio) if hypertension or hypokalaemia
Sex steroids	DHEA-S (dehydroepiandrosterone sulphate) etc. only if ACC suspected
Bilateral	17-hydroxyprogesterone (CAH); exclude adrenal insufficiency

- **Exclude phaeochromocytoma before biopsy or surgery.** Biopsy only for suspected metastasis with known cancer.

ESE–ENSAT 2023 (European Society of Endocrinology–European Network for the Study of Adrenal Tumors)

- Homogeneous **≤10 HU: benign, no further imaging whatever the size**; metanephrines may be omitted.
- MACS: screen for hypertension, diabetes and osteoporosis; surgery individualised.

Management

- Surgery: functioning tumour, suspicious imaging, >4 cm in fit patients, growth ≥5 mm and >20%.

Diagram to draw: imaging and biochemistry algorithm above.

> **Key points to remember:**
> - ≤10 HU = lipid-rich adenoma.
> - APW >60%, RPW >40%.
> - 1-mg DST: >1.8 µg/dL = MACS.
> - Rule out phaeochromocytoma before biopsy.

Rapid Revision Sheet – 2022-Summer Paper 4

All the key points from this paper in one place. Read this in the last 30 minutes before the exam.

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- AuroLase gold nanoshells: photothermal focal therapy.

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- Four-member Board; no transplant-team member.
- Two tests ≥6 h apart; Form 10.
- Apnoea test: no breathing at $PaCO_2$ ≥60 mmHg.

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- Semirigid: two channels; below the iliac vessels.
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